

ULMS Test Plan

1.1 Testing Objectives

System purpose: The Umoja Learning Management System (ULMS) WinForms application supports login authentication, student registration, course enrolment, marks capture, report generation and logout/navigation control.

Testing goals: verify that each module meets functional requirements, handles invalid input safely, calculates student averages correctly, prevents unauthorised access, and produces reports within an acceptable response time.

Scope of testing: login, dashboard navigation, student registration, course enrolment, marks capture, report generation, validation messages, basic security, usability, reliability and performance.

1.2 Testing Types Identification

Testing Type	ULMS Example	Method/Technique	Expected Outcome
Functional testing	Login validation	Positive and negative tests using valid and invalid credentials	Only valid admin/1234 credentials allow dashboard access
Input validation	Student registration and marks entry	Boundary value analysis and equivalence partitioning	Blank, non-numeric and out-of-range data rejected with clear messages
Integration testing	Dashboard to forms and course allocation	Navigation and workflow tests	Forms open correctly and return safely to dashboard
Performance testing	Report loading	Manual stopwatch/load timing over repeated runs	Report generation completes within 2 seconds
Usability testing	Navigation and clear buttons	Observation checklist	Labels, buttons and messages are understandable
Security testing	Authentication	Invalid login attempts and blank credentials	No access is granted unless username and password both match

1.3 Testing Environment Setup

Item	Specification
Hardware	Windows PC/laptop, minimum 8GB RAM, 64-bit processor
Operating System	Windows 10/11
Development Tool	Visual Studio 2022 or later with .NET Desktop Development workload
Framework	C# Windows Forms, .NET 8 Windows target
Test Data	Valid and invalid login credentials, valid/invalid student details, duplicate enrolment records, mark values 0-100, report filters
Evidence Tool	Windows Snipping Tool or Print Screen, ensuring taskbar clock/date is visible